

PSET 8: Multivariate distributions

Assignment Qs

- Name
- How long did this problem set take you (round to nearest hour)? 0-1 hour 2-3 hours 4-5 hours 5+ hours
- How difficult was this problem set? very easy 1 2 3 4 5 very challenging

Problem 1: Properties of a joint PDF

Continuous random variables X and Y have the following joint probability density function (PDF):¹

$$f_{XY}(x, y) = \begin{cases} kx^3y^2 & \text{where } 0 < x, y < 6 \\ 0 & \text{otherwise} \end{cases}$$

Note: $0 < x, y < 6$ means that both x and y are between 0 and 6; it does not mean that x is greater than 0 and y is less than 6.

- Find k .
- Find the marginal PDF of X , $f_X(x)$.
- Find the marginal PDF of Y , $f_Y(y)$.
- Find $\Pr(X < 2, Y > 4)$.

Problem 2: Calculating conditional PDF

Let $f(x, y) = 18x^2y^2$ for $0 \leq x \leq y \leq 1$.²

- Sketch the region on which $f(x, y)$ is positive.
- Find $f(x|y)$.
- Are X and Y independent? Explain your reasoning, paying attention to the region on which $f(x, y)$ is defined.

¹Inspired by Grimmer HW12.1

²Inspired by Grimmer HW12.4

Problem 3: Moments and independence

Continuous random variables W and Z have the following joint probability density function (PDF):³

$$f_{WZ}(w, z) = \begin{cases} kw^2z^3 & \text{where } 0 < w, z < 2 \\ 0 & \text{otherwise} \end{cases}$$

- Find k .
- Find $E[W]$.
- Find $Var(W)$.
- Find $Cov(W, Z)$.
- Are W and Z independent? Explain your reasoning using mathematical concepts from the course.
- What is the PDF of W conditional on Z , $f_{W|Z}(w|z)$?

Problem 4: Properties of joint random variables

Suppose the following:⁴

- $E[D] = 8$
 - $E[F] = 4$
 - $E[DF] = 10$
 - $Var(D) = 30$
 - $Var(F) = 60$
- What is $Cov(D, F)$?
 - What is the correlation between D and F ?
 - What is $Var(D + F)$?
 - What is $Var(D - F)$?
 - Suppose you multiplied F by 2 to generate a new variable, H . What is $Cov(D, H)$?
 - What is $Cor(D, H)$? How does this compare to your answer to part (b) of this question?
 - Suppose instead that $Var(D) = 40$. How would this change $Cor(D, F)$?

Problem 5: Continuous Bayes' theorem

Previously, we used Bayes' theorem to link the conditional probability of discrete events A given B to the probability of B given A . There is an analogous Bayes' theorem that relates the conditional densities of random variables X and θ :⁵

$$f(\theta | X) = \frac{f(X | \theta)f(\theta)}{\int f(X | \theta)f(\theta) d\theta}$$

³Inspired by Grimmer HW12.1

⁴Inspired by Grimmer HW12.3

⁵Inspired by Grimmer HW12.5

- a. Return to the joint PDF from Problem 2. Using the conditional density $f(x|y)$ and the marginal $f_Y(y)$ you worked with there, apply Bayes' theorem to find $f(y|x)$. Here y plays the role of θ and x plays the role of the observed data X . Be careful with the limits of integration in the denominator.
- b. Now compute $f(y|x)$ directly as $f(x, y)/f_X(x)$ and confirm that it matches your answer to part (a). What familiar quantity does the denominator of Bayes' theorem turn out to be here?

AI and Resources statement

Please list (in detail) all resources you used for this assignment. If you worked with people, list them here as well. It is not enough to say that you used a resource for help, you need to be specific on the link and *how* it was helpful. W/R/T gen AI tools (including GPT, Claude, etc.) you cannot use them to do work on your behalf—you cannot put in any of the questions, etc. You can ask for help on logic / sample problems. If you do use GPT or other AI tools, you need to provide a link to your chat transcript. Any suspected academic integrity violations will be immediately reported to the Dean of Students.

(Optional — complete elsewhere) Submission of practice questions

Submit practice questions for the final exam here: <https://forms.gle/sskPerZjthPYPbVt7> Note that we need at least 10 people to submit before there's enough to circulate!